①

$r = .55\,cm$ $LHR = 300\ W/cm$ $K_F = .025\ W/cm\text{-}K$

$E = 200\ GPa$ $\nu = .35$ $\alpha = 10\times10^{-6}\ 1/K$

$\boxed{93}$

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$$\sigma_r(\eta) = -\sigma^*(1-\eta^2) \qquad \sigma^* = \frac{\alpha E(T_0 - T_S)}{4(1-\nu)}$$

$$\sigma_\theta(\eta) = -\sigma^*(1-3\eta^2)$$

$$\sigma_z(\eta) = -2\sigma^*(1-2\eta^2) \qquad T_0 - T_S = \frac{LHR}{4\pi K}$$

$$T_0 - T_S = \frac{300}{4\pi(.025)} = 954.93$$

$$\sigma^* = \frac{10\times10^{-6}\cdot 200\cdot(954.93)}{4(1-.35)} \approx .734 \Rightarrow 734\ MPa$$

a.

$$\sigma_r = -\overset{734}{\cancel{954.93}}(1-1^2) = 0$$

$$\sigma_\theta = -\overset{734}{\cancel{954.93}}(1-3(1)^2) = \overset{+1468}{\cancel{1909.86}}\ MPa$$

$$\sigma_z = -2\cdot\overset{734}{\cancel{954.93}}(1-2(1)^2) = \underset{+1469}{\cancel{1909.86}}\ MPa$$

b. $\sigma_{fracture} = 175\ MPa$

$$175\ MPa = -\overset{734}{\cancel{954.93}}(1-3\eta^2)$$

$$\frac{175}{734} = 1-3\eta^2 \Rightarrow 3\eta^2 = .88$$

$$\eta^2 = \frac{.88}{3}$$

$$\eta = \sqrt{.29}$$

$$\boxed{\eta = .64}$$

(2) $P = 15 \text{ MPa}$

$\bar{R}_c = .55\text{cm} \quad t_c = .05\text{cm}$

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a.

$$\sigma_\theta = \frac{PR}{\delta} \Rightarrow \frac{15 \cdot .55}{.05} = 165 \text{ MPa}$$

$$\sigma_z = \frac{PR}{2\delta} \Rightarrow \frac{15 \cdot .55}{2 \cdot .05} = 82.5 \text{ MPa}$$

$$\sigma_r = -\frac{P}{2} \Rightarrow \frac{-15}{2} = -7.5 \text{ MPa}$$

b.

$$\sigma_r = -P\frac{(R_o/r)^2 - 1}{(R_o/R_i)^2 - 1}$$

$$\sigma_\theta = P\frac{(R_o/r)^2 + 1}{(R_o/R_i)^2 - 1}$$

$$\sigma_z = P\frac{1}{(R_o/R_i)^2 - 1}$$

~~$\frac{R_o + R_i}{2} = .55$~~

~~$R_o + R_i = 1.1$~~

~~$R_o = 1.1 - R_i$~~

~~$R_o - R_i = .05\text{cm}$~~

$R_o = R_i + .05\text{cm}$

$.05/2 = .025$

$R_o = .55 + .025$
$= .575$

$R_i = .55 - .025 = .525$

$r = R_i$

$$\sigma_r = -15\frac{\left(\frac{.575}{.525}\right)^2 - 1}{\left(\frac{.575}{.525}\right)^2 - 1} = -15 \text{ MPa}$$

$$\sigma_\theta = 15\frac{\left(\frac{.575}{.525}\right)^2 + 1}{\left(\frac{.575}{.525}\right)^2 - 1} \Rightarrow 15\,\frac{2.2}{.2} = 165 \text{ MPa}$$

$$\sigma_z = 15\frac{1}{\left(\frac{.575}{.525}\right)^2 - 1} \Rightarrow 15\,\frac{1}{.2} = 75 \text{ MPa}$$

③ $R_f = .48\text{ cm}$ $t_{gap} = .008\text{ cm}$ $T_{co} = 550\text{ K}$ $t_{clad} = .08\text{ cm}$

$K_{fuel} = .03\text{ W/cm-K}$ $K_{gap} = .003\text{ W/cm-K}$ $K_{clad} = .15\text{ W/cm-K}$

$LHR = 250\text{ W/cm}$ $\alpha_c = 10\times10^{-6}\text{ 1/K}$, $\alpha_f = 14\times10^{-6}\text{ 1/K}$ $T_{ref} = 300\text{ K}$

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$$\Delta\delta_{gap} = \bar{R}_c\,\alpha_c(\bar{T}_c - T_{ref}) - R_f\,\alpha_f(\bar{T}_f - T_{ref})$$

$$\bar{R}_c = \frac{R_i + R_o}{2} = \text{[illegible]}$$ $R_i = .48 + .008 = .488$

$R_o = .488 + .08 = .568$

$$\bar{R}_c = \frac{.488 + .568}{2} = .528$$

$$h_{clad} = \frac{K_{clad}}{t_{clad}} = \frac{.15}{.08} = 1.875 \quad h_{gap} = \frac{K_{gap}}{t_{gap}} = \frac{.003}{.008} = .375$$

$$T_{ic} - T_{oc} = \frac{LHR}{2\pi R_f h_{clad}} \Rightarrow T_{ic} = \frac{LHR}{2\pi R_f h_{clad}} + T_{oc} \Rightarrow \frac{250}{2\cdot\pi\cdot .48\cdot 1.875} + 550$$

$$T_{ic} = 594.21\text{ K}$$

$$\bar{T}_c = \frac{T_{oc} + T_{ic}}{2} = \frac{550 + 594.21}{2} = 572.1$$

$$T_s = \frac{LHR}{2\pi R_f h_{gap}} + T_{ic} \Rightarrow \frac{250}{2\cdot\pi\cdot .48\cdot .375} + 594.21 = 815.26$$

$$T_o = \frac{LHR}{4\pi K} + T_s = \frac{250}{4\cdot\pi\cdot .03} + 815.26 = 1478.41$$

$$\bar{T}_F = \frac{T_o + T_s}{2} \Rightarrow \frac{1478.41 + 815.26}{2} = 1146.83$$

$$\Delta\delta_{gap} = (.528)(10\times10^{-6})(572.1 - 300) - (.48)(14\times10^{-6})(1146.83 - 300)$$

$$\Delta\delta_{gap} = .00144 - .0057 = \boxed{-.00426\text{ cm}}$$

(4)

$\Delta T = 50 \quad \alpha_c = 15\times10^{-6} \quad E = 100\ GPa \quad \nu = .34$

$t_c = .06cm \quad R_i = .55cm$

6/10

hoop and radial at inner radius

$$\sigma_r = \frac{1}{2}\Delta T\frac{\alpha E}{1-\nu}\left(\frac{r}{r_i}-1\right)\left(1-\frac{R_i}{\delta}\left(\frac{r}{R_i}-1\right)\right)$$

$r = R_i$

$$\sigma_\theta = \frac{1}{2}\Delta T\frac{\alpha E}{1-\nu}\left(1-2\frac{R_i}{\delta}\left(\frac{r}{R_i}-1\right)\right)$$

$$\sigma_r = \frac{1}{2}\Delta T\frac{\alpha E}{1-\nu}\left(\cancel{1-1}\right)\left(1-\frac{.55}{.06}\left(\cancel{1-1}\right)\right) = 0\ MPa$$

$$\sigma_\theta = \frac{1}{2}\Delta T\frac{\alpha E}{1-\nu}\left(1-2\frac{R_i}{\delta}\cancel{(1-1)}\right) \Rightarrow \frac{1}{2}(50)\frac{15\times10^{-6}\cdot 100}{1-.34}(1) = 56.8\ MPa$$

⑤

$\dot{F} = 6\times10^{13}$ fiss/cm^3-s $\quad \Delta\rho_0 = .015 \quad B_0 = 5$ MWD/kg U

$\rho_{UO_2} = 10.97$ g/cc $\quad t = 10$ days $\quad C_0 = 1$

1 day 24 hrs 60 min 60 secs
1 day 1 hr 1 min

$$\varepsilon_D = \Delta\rho_0 \left(e^{\frac{B \ln .01}{C_0 B_0}} - 1\right)$$

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$$B = \frac{\dot{F} t}{N_U}$$

237?

$$N_U = 10.97\,\text{g/cc} \cdot \frac{1}{237.91} \cdot 6.022\times10^{23} = 2.78\times10^{22}$$

$$B = \frac{6\times10^{13} \cdot 10 \cdot 24 \cdot 60 \cdot 60}{2.78\times10^{22}} = .0019$$

$$B_0 = 5/950 = .005$$

$$\varepsilon_D = .015\left(e^{\frac{.0019 \ln .01}{.005}} - 1\right) = -.012$$

- only wrong mass

(6) Elasticity is the ability of a material to resist ~~change in atomic configuration~~ deformation due to an applied ~~stress~~ load

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plasticity is the ability of a material to deform under an applied load

-mostly, but needed more...

(7) Strain hardening is the increase in yield strength of a material due to plastic deformation. Strain hardening is caused when a material under load exceeds its yield stress and the load is subsequently removed. The new yield stress will be at the point of stress where maximum strain was previously reached.

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-but why?

(8)
- calculate temperature profile
- determine stresses due to mechanical and temperature changes
- determine strain from mechanical and temperature changes

2/3

-gap?

(9) The individual grains of the powder are compacted together through heat and pressure. As they are formed they change from separate units into a combined pellet when the previously separate grains are now connected via grain boundaries

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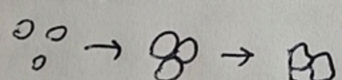

(10) The concept is based on using fundamental mechanical behavior of a system such as gas bubbles aggregating on a boundary or grain growth to calculate properties such as thermal conductivity. This is beneficial because there is only so much empirical data and it has its limitations for untested applications. By using a mechanistic modeling approach, the fundamental behavior of material can be used to accurately model applications that may not have adequate empirical data.

-not just mechanical

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(11) The microstructure is the structure of grains at 25x magnification. Cold working is used to deform the microstructure below the temperature when grains will reorient. This deforms the structure, introducing more ~~[illegible]~~ defects, making the material stronger.

-below recrystallization

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(12) Diffusion occurs more quickly along grain boundaries as there is more space for the diffusing atoms to go, and they will be able to settle at a lower energy state. This effect is amplified at lower temperatures.

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(13) Grain growth is driven by reducing the grain boundary energy. Grain growth can be impeded by solute atoms.

6/6

-Wanted more, but yes

(14) Fuel densifies because after manufacturing it is not fully dense and has pores within the pellet. Once operation begins and fissions happen in significant numbers, the energy from the fissions causes interstitial defects to ~~[illegible]~~ move within the lattice, reducing pore size and causing movement to the boundaries. This reduction in pore sizes causes the volume of the fuel to decrease for about the first 5 MWD of burnup.

5/6

-mostly